



Rating-based regulations and rating inflation: New evidence from quantitative easing programs[☆]

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ABSTRACT

We show that S&P upgraded abnormally more BBB+ firms to the A- category when rating-based capital requirements caused a strong demand from insurance companies for A- bonds during the Maturity Extension Program and the third round of quantitative easing. Meanwhile, there was no sign of improvement in the financial performance of the upgraded firms. These firms did not expand investments after the upgrades, but increased payouts to investors. Notably, the BBB+ firms with higher interest burdens and more bond issuance previously were more likely to be upgraded to the A- category. Additionally, compared to Egan-Jones Ratings, an investor-paid rating agency, S&P lowered its bar for A- ratings in 2013. Overall, we provide new evidence on the impact of rating-based regulations on credit ratings.

1. Introduction

The quantitative easing (QE) programs, launched by the Federal Reserve (the Fed) in response to the financial crisis of 2008–2009, significantly reduced the long-term interest rates in the post-crisis period. Among the multiple rounds of QEs, the Maturity Extension Program (MEP) and the third round of QE (QE3) effectively lowered the long-term interest rates for Treasury securities to historically low levels.

As a result, yield-reaching institutional investors subject to rating-based capital requirements (e.g., insurance companies) substituted corporate bonds with high credit ratings for Treasury securities (Foley-Fisher et al., 2016).¹ The demand for highly-rated corporate bonds was particularly strong for A- bonds because A- bonds offered the highest yields among corporate bonds subject to the same lowest capital requirement (Category 1).² This sharp rise in the demand for A- corporate bonds significantly lowered the yields of those bonds and thus the borrowing cost of

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¹ When an insurance company purchases a corporate bond, it must obtain a rating from at least one Nationally Recognized Statistical Rating Organization (NRSRO). The insurer then assigns the bond to the six-tier scale of the National Association of Insurance Commissioners (NAIC) based on that rating. Critically, the NAIC does not require insurers to average ratings across multiple agencies or to use the lowest (most conservative) rating. An insurer may use the most favorable rating from any NRSRO, provided the rating agency is recognized and the insurer applies its methodology consistently. Insurance companies are not required to file every bond purchase with the NAIC for pre-approval. Instead, they self-report their bond holdings and the corresponding NAIC designations in their annual statutory financial statements. The NAIC's Securities Valuation Office (SVO) conducts post-hoc reviews of these filings to verify that insurers have correctly mapped credit ratings to the NAIC categories. Insurers are required to use the most current credit rating available as of their financial reporting date. If a bond's rating is upgraded or downgraded during the year, the insurer must update the NAIC designation accordingly as of the effective date of the rating change.

² We present the rating-based capital requirements in Appendix A. Category 1 bonds include bonds rated AAA through A-. For every \$100 investment in Category 1 bonds, life insurance companies are required to maintain \$0.30 of equity capital (after tax) which is considerably lower than the requirement of 1.00% (after tax) for Category 2 bonds (i.e., bonds rated BBB+, BBB, or BBB-). Other types of insurance companies face similar capital requirements. For a \$100 million bond investment, an insurer would need to hold \$300,000 in capital if the bond is rated A-, but \$1,000,000 if it is rated BBB+ — a difference of \$700,000. This difference gives insurers a strong preference for A- bonds over BBB+ bonds when seeking substitutes for low-yield Treasury securities.

the issuers with A- ratings, which increased the regulatory advantage of A- ratings.³ As shown in Fig. 1, the spread between A- and BBB+ bonds rose significantly during this period, making BBB+ firms with unmet financing needs during the crisis growingly eager to obtain higher ratings. In this study, we examine whether the strong investor demand for A- bonds induced by rating-based capital requirements in the MEP and QE3 periods caused rating inflation around the A- rating.

On the one hand, rating-based regulations often create incentives for issuers just below the regulation thresholds to put pressure on credit rating agencies (CRAs) to obtain higher grades. A number of studies have provided evidence that issuer-paid CRAs tend to cater to their clients (Jiang et al., 2012; Cornaggia and Cornaggia 2013; Xia, 2014; Kraft, 2015), and such conflicts of interest appear to be more pronounced around the investment-grade threshold (Jiang et al., 2012; Bruno et al., 2016; Behr et al., 2018). This is because the regulatory advantage of highly rated securities can generate more economic benefits for issuers to share with CRAs (Opp et al., 2013). In light of these studies, it is reasonable to predict that an issuer-paid CRA will be more likely to issue A- ratings when the strong demand for A- bonds increases issuers' benefits from upgrades.

On the other hand, in response to the increasing criticism from the public and regulators, CRAs have become more stringent in assigning ratings (Baghai et al., 2014), which can weaken the impact of rating-based regulations on corporate credit ratings. Although the quality of CRAs' ratings on financial instruments was criticized for failing to portray the default risks before the financial crisis of 2008–2009, the deterioration of rating quality did not spill over to corporate ratings. Actually, corporate ratings have improved since this crisis, suggesting that the CRAs reacted positively to public critics and regulatory pressure (deHaan, 2017). In addition, the Congress passed the Dodd-Frank Act in 2010 to enhance the regulations over CRAs, which has limited CRAs' flexibility in rating issuance (Dimitrov et al., 2015).⁴ As such, it is possible that the changes in investor demand induced by the unconventional monetary policies did not result in rating inflation around the A- threshold.

To empirically address our research question, we focus on the upgrades of BBB+ firms by S&P, an issuer-paid rating agency, when there was a high demand for A- debt securities (i.e., 2012–2013). If the rating-based regulations discussed previously indeed affected rating outcomes in the high-demand period, we expect to observe more firms being upgraded from BBB+ to A- by S&P during this period.⁵

We first examine the abnormality of the upgrades of BBB+ firms by

S&P during the high-demand period, the years when Treasury yields delved to historical lows and the demand for A- bonds most likely peaked.⁶ We focus on the upgrades of BBB+ firms because firms are usually upgraded by one notch. Analyzing firms' annual long-term ratings for 2003–2016, we find the difference in the upgrade likelihood between BBB+ and other firms in 2012–2013 to be significantly higher than that in other years. In other words, compared to firms with other ratings, BBB+ firms were more likely to be upgraded in this period than in other years. Acknowledging that the likelihood of upgrades can vary with the ratings, we then focus on BBB+ firms and firms with ratings close to BBB+ (i.e., A-, BBB, and BBB-). We still observe a spike in the likelihood of upgrades for BBB+ firms during the high-demand period. Furthermore, these results continue to hold when we restrict the sample to only BBB+ firms.

To better detect the abnormality in rating upgrades, we narrow our focus to the yearly rating patterns for BBB+ firms. We find that BBB+ firms were abnormally more likely to be upgraded in 2013, the latter half of the high-demand period, than in other years. This could be attributed to the substantial increase in bond issuance by A- firms in 2012 (Foley-Fisher et al., 2016) which might have weakened CRAs' incentives to cater to other clients.

To substantiate our contention that the abnormal upgrades are driven by the shock to the regulatory advantage of A- securities, we also examine the patterns of upgrades across firms with ratings other than BBB+. Specifically, we perform the upgrade tests on firms with ratings from A- through B- separately (except for BBB+), but do not observe any abnormal pattern in their likelihood of being upgraded. These results suggest that the abnormal upgrades are limited to the rating category affected the most by the unconventional monetary policies during the high-demand period.

To further consolidate our conclusion, we compare the expected rating scores of the upgraded BBB+ firms and other BBB+ or A- firms. Following Baghai et al. (2014), we calculate the expected rating scores for each firm. We do not find significant differences in the expected rating scores between the upgraded and non-upgraded BBB+ firms, but such differences are significant between the upgraded BBB+ firms and other A- firms. These results suggest that the creditworthiness of the upgraded BBB+ firms is lower than what would be considered justifiable by normal standards.

We also examine the changes in the financial indicators valued by CRAs for firms rated BBB+ at the end of 2012. The results show that the average financial indicators of the upgraded firms did not exhibit signs of significant improvement around the upgraded year. Additionally, the results indicate that the interest expenses of the upgraded BBB+ firms decreased after the upgrades, suggesting the upgrades effectively reduced the interest expenses of those upgraded firms. We also find that the upgraded BBB+ firms did not significantly change their investments, but increased their payouts to investors after the upgrades.

To test the prediction that the pattern of the abnormal upgrades is related to the potential economic benefits to firms, we perform the main regressions again with the measures of interest burden included using the sample of BBB+ firms. The results indicate that firms with higher interest burdens (i.e., measured by interest to debt ratio, the reverse of interest coverage ratio, and total debt to EBITDA ratio) were significantly more likely to be upgraded in 2013. The findings support the conjecture that firms with higher interest burdens, which stand to gain more from obtaining an A- rating, have a stronger incentive to seek rating upgrades when the regulatory advantage of A- debt securities is

³ See Appendix B for more detailed introduction of the institutional background.

⁴ Subtitle C of Title IX of the Dodd-Frank Act lays out new rules and initiatives regarding the regulation of credit rating agencies. The subtitle mandates the creation of an Office of Credit Ratings (OCR) by the Securities Exchange Commission to oversee nationally recognized statistical rating organizations (NRSROs). To address the problem of inaccurate ratings, the new regulations in Dodd-Frank Act require NRSROs to adopt an internal control system and corporate governance structure to ensure adherence to policies, procedures, and methodologies in credit rating. In addition, the new regulations also specify mechanisms to reduce conflict of interests arising from sales, marketing, and other unwarranted considerations.

⁵ We focus on the upgrades of firms rather than bonds because BBB+ firms without outstanding bonds but planning to issue bonds in the future are also incentivized to obtain higher ratings when the regulatory advantage of the A- rating is high. Solely looking at ratings on bonds would overlook these potential issuers. Additionally, by focusing on the ratings on firms rather than the ratings on new bond issuance, we can avoid the identification challenges posed by self-selection in bond issuance. Firms that qualify for an A- rating were more likely to issue bonds when the strong demand for A- bonds drives down the yields of those bonds (Foley-Fisher et al. 2016).

⁶ The Treasury rates dropped to a historically low level in 2012, and the Fed's holding of long-term Treasury securities peaked in 2013 (as shown in Figure 2 and Figure 3).

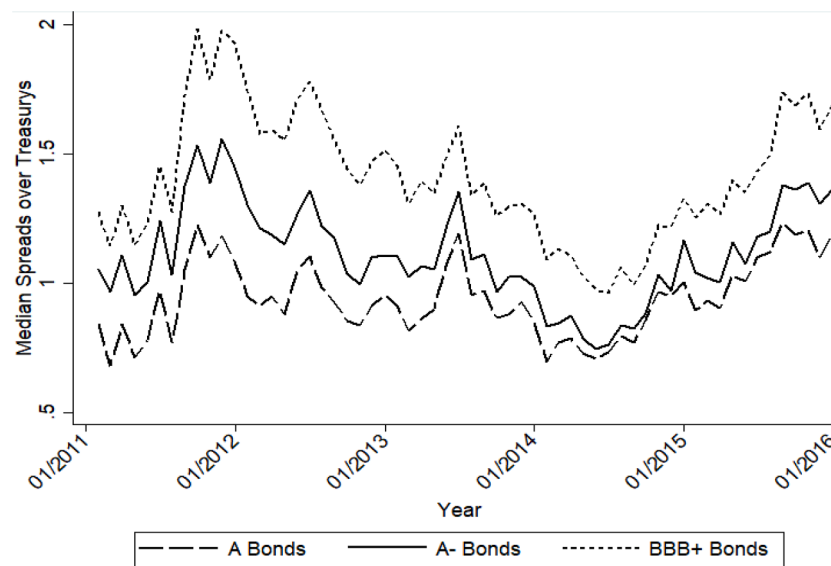


Fig. 1. Yield spreads of A, A- and BBB+ bonds.

larger.

We also examine the rating agency's incentive to issue abnormally more upgrades. Both [Jiang et al. \(2012\)](#) and [Cornaggia and Cornaggia \(2013\)](#) provide evidence that CRAs' revenue model can bias their rating decisions and issue favorable ratings for their clients. Therefore, we expect the upgrade likelihood to be higher for firms that are the rating agency's active clients and still have debt capacity. Our results indicate that, BBB+ firms that issued more bonds in the prior three years but still retained debt capacity were more likely to be upgraded in 2013 than other firms, which lends further support to our contention. We also find that these firms issued more bonds in the subsequent year.

To strengthen our argument, we also use the ratings of Egan-Jones Ratings (EJR), an investor-paid CRA, as an alternative control group. The results suggest that EJR assigned lower ratings to firms rated A- by S&P in 2013. However, this pattern is not observed for ratings other than A-. The findings support the notion that S&P is responsive to the increased regulatory advantage of A- rating category during the high-demand period. Additionally, our results are robust to several alternative research designs.

This study deepens the understanding of how rating-based regulations shape credit rating outcomes. By creating regulatory advantages at specific rating thresholds, these regulations incentivize firms near the thresholds to seek rating upgrades and bias credit ratings ([Kisgen and Strahan, 2010](#); [Bongaerts et al., 2012](#)). Prior studies have documented rating inflation resulting from rating-based regulations around the investment-grade cutoff ([Jiang et al., 2012](#); [Bruno et al., 2016](#); [Behr et al., 2018](#); [Stanton and Wallace, 2018](#)). Our study contributes to this line of research by highlighting the time-varying nature of the impact of rating-based regulations. We show that as the regulatory advantage of a rating increases with changing economic conditions, rating inflation around this rating can become more profound as well. Specifically, we provide evidence that when investors' demand for A- bonds spiked during MEP and QE3, CRAs upgrade abnormally more firms to the A- category, a regulatory threshold that has not been emphasized by previous research. Our results suggest that extant research may underestimate the likelihood of rating inflation when economic conditions change significantly, such as during periods of unconventional monetary policies. Our findings also underscore the need for dynamic oversight

mechanisms that account for how changes in economic conditions amplify distortions in CRAs' behavior.⁷

Our study is also related to research on the independence of CRAs. While the literature has firmly established the importance of credit ratings in the financial market ([Grier and Katz, 1976](#); [Ingram et al., 1983](#); [Ederington et al., 1987](#); [Livingston et al., 2010](#); [Kliger and Sarig, 2000](#); [Tang, 2009](#); [Cornaggia et al., 2018](#)), prior studies also find that CRAs adjust ratings for reasons unrelated to the creditworthiness of firms. The independence of CRAs and the quality of their ratings have been frequently questioned by the public and regulators over the past two decades. After the financial crisis of 2008–2009, the US Congress passed the Dodd-Frank Act to strengthen the regulatory scrutiny over CRAs and improve the quality of ratings issued by national CRAs. While in response to public critics and regulatory scrutiny, CRAs have become more conservative when issuing ratings, especially after the passage of the Dodd-Frank Act ([Dimitrov et al., 2015](#)), our results highlight the fact that CRAs still retain significant flexibility in issuing ratings. In particular, it is possible that the inflation in ratings we identify is one of the consequences of S&P's review of their rating methodology, which is consistent with the finding in [Beatty et al. \(2019\)](#) that CRAs use their adjustment to rating methodology for their own benefit.

Furthermore, our study is related to recent research on the rating inflations during the quantitative easing programs worldwide. [Acharya et al. \(2021\)](#) show that CRAs delay the downgrades of BBB-rated firms that engage in M&As to the high-yield status during the large-scale assets purchase programs by the Fed. Compared to [Acharya et al. \(2021\)](#) that examine the rating patterns of BBB-rated firms over a long period, our study exploits an exogenous increase in the demand of institutional investors subject to rating-based capital requirements for A- bonds in a relatively short period to show the abnormal upgrades of A- firms. [Abidi](#)

⁷ Instead of static rules, a dynamic oversight mechanism could require more frequent and targeted reviews of rating migrations around key thresholds (e.g., A- or BBB-) during periods of unconventional monetary policy or other macroeconomic shocks that significantly alter investor demand. For example, during large-scale asset purchase programs or periods of macroeconomic shocks, regulators could first identify key rating thresholds and then mandate that CRAs provide ex-post justifications for all upgrades around the thresholds, allowing the SEC's Office of Credit Ratings to more easily identify potential rating inflation.

et al. (2023) examine the impact of the European Central Bank's corporate sector purchase program (CSPP)-a form of quantitative easing program-on credit ratings. Their key finding is that rating activities are significantly concentrated around bonds near the CSPP eligibility threshold of BBB-. Our work differs from Abidi et al. (2023) in that we focus on the impact of specific rating-based capital requirements imposed on institutional investors rather than that of rating-based monetary policies.⁸

2. Upgrades in the high-demand period

2.1. The sample

We collect data on S&P's credit ratings and firms' financial characteristics for the period of 2003–2016 from Compustat.⁹ Firm-year observations without necessary information are excluded from the sample. Financial firms and firms in regulated industries are also excluded. We further eliminate the rare cases in which firms' ratings move by more than one notch.¹⁰ The final sample includes 15,185 observations. As reported in Panel A of Table 1, the firm-year observations are quite evenly distributed across years.

Panel B of Table 1 reports the yearly distribution of the sample by rating category. According to S&P's rating classification, our sample includes firms rated across 20 categories from AAA through CC.¹¹ The number of A- firms in our sample decreased over the financial crisis of 2007–2009 and started to bounce back in 2010. There was a significant jump in the number of A- firms in 2013, the latter half of the period when investors' demand for A- bonds was high.¹² The number of A-firms stayed stable afterward. As for the number of A firms, the pattern is different. Over years, the number of A firms decreased gradually and did not appear to bounce back. The pattern in the number of BBB+ firms is similar to that of A- firms, but the number of BBB+ did not show any

significant increase after its bouncing back in 2011 until 2016.

Panel C reports the yearly upgrades by firms' prior ratings (i.e., ratings prior to the upgrades). There were 10 and 24 firms upgraded from BBB+ to A- in 2012 and 2013 respectively. The number of upgrades of BBB+ firms in 2013 was significantly more than those in other years. In the three years prior to 2013, the total amounts of bond issuance of these upgraded BBB+ firms was 16.84% of the total bond issuance of all the BBB+ firms at the end of 2012. We observe no such significant upgrades in other ratings.

We focus on the abnormal upgrades of firm-level ratings instead of those of new issues because the sample of new issues is largely affected by firms' willingness to issue bonds. For example, in 2012, many A-firms issue bonds to take advantage of the sharp drop in cost of debt. The estimated average quality of A- firms that issue new bonds can be largely affected by the issuers in the sample. We also do not use the sample of outstanding bonds, which is because the sample of outstanding bonds does not include potential issuers.

2.2. Abnormal upgrades

2.2.1. Upgrades in 2012–2013

To test for the abnormal upgrades in the high-demand period when the regulatory advantage of A- securities was high, we first examine whether the upgrades in the high-demand period are different from those in other years. More specifically, we investigate whether BBB+ firms were more likely to be upgraded in 2012–2013.

We first conduct the tests using the full sample. The tests follow a difference-in-difference framework. The first-level difference is the difference between the tendency to be upgraded for firms with a BBB+ rating and that for firms with other ratings in the same year. The treatment effect is then reflected in the difference between the first-level difference for the high-demand period and those for other years. The logistic model is specified as follows.

$$\text{Prob}(UP_{it} = 1) = \psi(\alpha_{HD} + \beta_{HD}X_{it} + \tau_{HD}BBB_{+it} + \gamma_{HD}HighDemand_t + \theta_{HD}HighDemand_t \times BBB_{+it} + \varepsilon_{it}) \quad (1)$$

UP is an indicator variable equal to one if a firm i is upgraded in the current year t , and zero otherwise. BBB_{+} is an indicator variable equal to one if a firm i was rated BBB+ in year $t-1$ and zero otherwise. The variable $HighDemand$ is an indicator of the high-demand period (i.e., 2012–2013) when the monetary easing increased the demand for A-bonds. The parameter θ_{HD} captures the difference in the likelihood of being upgraded for BBB+ firms in the high-demand period and that of the other firms in the same period, conditional on the normal difference between the upgrade likelihood for BBB+ firms and firms with other ratings. As discussed in the previous section, the demand for A- bonds was particularly high from 2012 to 2013 because of the demand shock associated with the Fed's open market operations. Therefore, we expect θ_{HD} to be significantly positive (i.e., the BBB+ firms are more likely to be upgraded relative to other firms in the high-demand period).

The control variables X_{it} in Model (1) include all the financial variables employed by Baghai et al. (2014) to estimate firm ratings: the sum of long-term and short-term debt divided by total assets (*Leverage*), convertible debt divided by total assets (*Convert*), rental payments divided by total assets (*Rent*), cash and marketable securities divided by total assets (*Cash*), the sum of long-term and short-term debt divided by EBITDA (*Debt*), interest payments divided by EBITDA (*IntCov*), EBITDA divided by sales (*Profit*), the logarithm of the book value of assets (*Size*), net property, plant, and equipment divided by total assets (*PPE*), capital expenditures divided by total assets (*Capital*), and the standard deviation of *Profit* over the past five years (*ProVol*).¹³ As we focus on firms'

⁸ We differ from Abidi et al. (2023) in two additional aspects. First, Abidi et al. (2023) focus on bonds near the BBB- threshold before and after the CSPP and examine rating activities around the CSPP eligibility threshold, while our empirical analysis focuses on BBB+ firms and employs a difference-in-difference framework to examine the likelihood of upgrades during the high-demand period. Second, Abidi et al. (2023) emphasize the need for regulators to understand the impact of rating-based unconventional monetary policies on credit ratings, suggesting that such regulatory policies can create incentives for rating inflation. Our study instead highlights the joint impact of unconventional monetary policies and rating-based capital market regulations, showing that even non-rating-based unconventional monetary policies can still induce rating inflation due to rating-based regulations.

⁹ The sample period covers only the post-SOX era. Compustat no longer provides S&P's credit ratings at the firm level after February 2017, so our sample period does not extend beyond 2016.

¹⁰ Firms that are upgraded or downgraded by more than one notch each time usually have more extreme changes in performance. Including these cases in our estimation may increase the likelihood for our results to be affected by outliers, so we exclude these cases from our sample. In our sample period, before we match any financial or return data, there were 654 cases with ratings upgraded by more than one notch, out of which 53 cases were upgraded to A-. There were only 3 firms that were upgraded by more than one notch to A- in 2013. Our conclusion remains unchanged if we do not exclude these cases.

¹¹ No firms rated C satisfy our sampling criteria, and thus our sample only includes firms rated CC or above.

¹² The jump in A- firms in 2013 was not caused by the recovery of firms that were previously downgraded during the financial crisis of 2007–2009. As we show later, out of the 24 firms that were upgraded to A- in 2013, only 4 were downgraded in the financial crisis. In our main tests, we include an indicator variable for firms that were downgraded in the financial crisis.

¹³ We define *IntCov* as interest expenses divided by EBITDA instead of the reverse to avoid extreme values that can affect the regression results.

Table 1
Distribution of observations.

Panel A: Yearly distribution														
Year	Observations								Percentages					
2003	1080								7.11					
2004	1145								7.54					
2005	1125								7.41					
2006	1112								7.32					
2007	1062								6.99					
2008	1029								6.78					
2009	1026								6.76					
2010	1039								6.84					
2011	1045								6.88					
2012	1096								7.22					
2013	1112								7.32					
2014	1147								7.55					
2015	1101								7.25					
2016	1066								7.02					
Total	15,185								100					
Panel B: Yearly distribution by ratings														
Ratings	2003	2004	2005	2006	2007	2008	2009	2010	2011	2012	2013	2014	2015	2016
AAA	9	7	7	8	8	6	5	6	7	6	6	6	5	3
AA+	2	3	2	3	2	2	2	1	1	1	1	2	1	3
AA	12	13	13	13	15	13	12	12	9	11	12	12	8	8
AA-	12	14	12	12	12	14	15	16	15	12	18	18	21	19
A+	33	31	33	29	28	23	24	22	27	29	26	26	23	24
A	73	76	70	66	72	67	68	63	59	60	59	62	54	55
A-	68	75	81	74	53	54	49	64	63	67	86	88	88	82
BBB+	82	100	109	96	99	88	92	94	113	118	106	113	116	125
BBB	145	163	145	140	125	139	147	144	147	158	167	159	158	141
BBB-	93	94	96	92	98	100	101	124	109	113	118	112	110	107
BB+	77	77	64	62	63	55	51	54	74	79	70	86	86	92
BB	89	95	109	102	92	84	91	88	89	91	94	101	100	99
BB-	135	144	132	151	140	123	109	102	94	96	109	111	101	113
B+	113	120	121	121	112	106	98	94	89	100	96	93	100	74
B	71	66	62	72	90	95	86	102	103	107	93	102	69	68
B-	35	46	42	47	37	49	60	41	34	39	36	43	44	31
CCC+	12	11	20	17	13	8	11	9	7	4	11	13	16	19
CCC	14	9	7	6	2	2	4	1	5	5	3	0	0	2
CCC-	3	0	0	1	0	0	0	2	0	0	1	0	1	1
CC	2	1	0	0	1	1	1	0	0	0	0	0	0	0
Total	1080	1145	1125	1112	1062	1029	1026	1039	1045	1096	1112	1147	1101	1066
Panel C: Yearly upgrades by prior ratings														
Ratings	2003	2004	2005	2006	2007	2008	2009	2010	2011	2012	2013	2014	2015	2016
AAA	0	0	0	0	0	0	0	0	0	0	0	0	0	0
AA+	0	0	0	0	0	0	0	1	0	0	0	0	0	0
AA	0	0	0	0	0	1	0	0	0	0	0	0	0	0
AA-	0	2	0	1	2	0	0	1	0	2	1	0	0	1
A+	0	0	0	1	2	1	0	2	0	1	5	0	2	0
A	2	3	6	2	2	4	0	3	1	1	4	1	1	0
A-	3	2	3	8	9	2	2	2	2	2	6	3	6	5
BBB+	3	1	11	4	3	6	2	10	6	10	24	5	9	6
BBB	5	7	16	10	14	7	6	20	19	11	12	13	6	20
BBB-	13	9	16	10	10	9	12	15	22	16	16	12	12	10
BB+	9	8	14	9	9	13	5	11	7	11	14	3	8	7
BB	10	7	6	10	13	16	3	16	25	14	11	19	15	8
BB-	14	12	25	17	14	17	12	14	23	13	20	22	10	10
B+	17	15	16	20	22	16	5	19	23	13	23	21	11	16
B	8	11	11	16	15	9	10	20	17	19	15	22	18	10
B-	2	5	6	6	10	8	8	27	10	12	5	8	4	4
CCC+	4	4	1	4	4	3	6	6	4	2	1	2	1	0
CCC	1	4	1	1	2	0	1	0	0	0	1	1	0	1
CCC-	1	1	0	0	0	0	0	0	1	0	0	0	0	0
CC	0	0	0	0	0	0	0	1	0	0	1	0	0	0
Total	92	91	132	119	131	112	72	168	160	127	159	132	103	98

upgrades, we also include the changes in these variables denoted as DZ , where Z is the control variables discussed above.

Following Purda (2007), we also control for the trends in firms'

performance. Specifically, we regress firms' total debt, long-term debt, ROA, and the reverse of the interest coverage ratio on time for the previous eight quarters ending by the end of the previous year.¹⁴ The

¹⁴ We thank one reviewer in our previous submission for recommending the approach proposed by Purda (2007).

Table 2
Summary statistics.

Variables	N	Mean	Std. Dev.	P25	Median	P75
<i>TrendTd</i>	15,185	0.275	1.864	-0.616	0.000	0.791
<i>TrendDlt</i>	15,185	0.299	1.885	-0.600	0.000	0.861
<i>TrendROA</i>	15,185	-0.029	0.413	-0.113	0.000	0.098
<i>TrendIntcov</i>	15,185	-0.097	1.808	-0.210	0.024	0.266
<i>Leverage</i>	15,185	0.367	0.218	0.226	0.328	0.457
<i>Convert</i>	15,185	0.014	0.047	0.000	0.000	0.000
<i>Rent</i>	15,185	0.018	0.027	0.004	0.009	0.018
<i>Cash</i>	15,185	0.083	0.092	0.017	0.053	0.115
<i>Debt</i>	15,185	3.295	3.717	1.567	2.729	4.341
<i>IntCov</i>	15,185	0.218	0.295	0.082	0.160	0.280
<i>Profit</i>	15,185	0.200	0.147	0.104	0.170	0.274
<i>Size</i>	15,185	8.490	1.440	7.463	8.396	9.457
<i>PPE</i>	15,185	0.370	0.258	0.140	0.312	0.589
<i>Capital</i>	15,185	0.057	0.053	0.023	0.042	0.072
<i>ProVol</i>	15,185	0.041	0.082	0.012	0.021	0.038
<i>Predown</i>	15,185	0.136	0.343	0.000	0.000	0.000
<i>DLeverage</i>	15,185	0.002	0.075	-0.030	-0.004	0.025
<i>DConvert</i>	15,185	-0.001	0.026	0.000	0.000	0.000
<i>DRent</i>	15,185	0.000	0.006	-0.001	0.000	0.001
<i>DCash</i>	15,185	0.000	0.050	-0.014	0.000	0.016
<i>DDebt</i>	15,185	-0.041	3.935	-0.478	-0.048	0.364
<i>DIntCov</i>	15,185	-0.005	0.295	-0.030	-0.004	0.018
<i>DProfit</i>	15,185	0.000	0.085	-0.012	0.001	0.015
<i>DSize</i>	15,185	0.054	0.189	-0.022	0.034	0.103
<i>DPPE</i>	15,185	-0.001	0.043	-0.014	0.000	0.013
<i>DCapital</i>	15,185	0.000	0.028	-0.007	0.000	0.007
<i>DProVol</i>	15,185	0.000	0.042	-0.004	0.000	0.003

trends in performance are the coefficients on time (*TrendTd*, *TrendDlt*, *TrendROA*, and *TrendIntcov*). Moreover, we also control for whether a firm was downgraded during the financial crisis of 2007–2009 (*Predown*) to exclude the possibility that the upgrade pattern for BBB+ firms in the high-demand period is driven by the massive downgrades of A- firms during the crisis.

The precise definitions of the control variables are presented in Appendix C. All the control variables except indicator variables are winsorized at the 1% level at each tail of their distribution. Table 2 reports the summary statistics of the control variables. The average trends for total and long-term debt (*TrendTd* and *TrendDlt*) are positive, which is consistent with the change in leverage (*dLeverage*) being positive on average. The trends for ROA and *IntCov* (*TrendROA* and *TrendIntcov*) are negative, which is consistent with the changes in firm profits (*dProfit*) being very small on average and the changes in *IntCov* (*dIntCov*) having a negative average value.

We do not control for any market-based measure of firm risk, such as probability of default calculated based on the Merton (1974)/KMV model, because a portion of bond issuers are not public firms. Including market-based probability of default can wipe out yearly upgrades that are meaningful for investors. But later robustness tests show that including the market-based risk measure does not change the results.

The regression results for the sample of all firms are reported in Column (1) in Panel A of Table 3. Standard errors are clustered at the firm level. The coefficient on the interaction term of the indicator variable for the 2012–2013 period and BBB+ (θ_{HD}) is positive and statistically significant, which is consistent with our expectation. The coefficient on BBB+ is significantly negative, which is primarily because the likelihood of firms being upgraded can differ across ratings, and firms with low ratings could be more likely to be upgraded. For example, it is reasonable to believe that firms rated BBB- are more likely to be upgraded than A firms. To avoid any concerns over the heterogeneous

nature of the likelihood of being upgraded across rating categories, we also perform the tests using a sample that only contains firms with A-, BBB+, BBB, and BBB- ratings. The sample includes 5923 firm-year observations. As reported in Column (2) in Panel A of Table 3, the results remain unchanged.¹⁵

Furthermore, we conduct the test using a sample of only firms rated BBB+ in the previous year. The test examines the likelihood of BBB+ firms being upgraded across years. The regression model is specified as follows.

$$Prob(UP_{it} = 1) = \psi(a'_{HD} + \beta'_{HD}X_{it} + \tau'_{HD}BBB_{+it} + \gamma'_{HD}HighDemand + \varepsilon'_{it}) \quad (2)$$

The regression results are reported in Column (3) in Panel A of Table 3. The sample includes 1424 firm-year observations. The pseudo R^2 rises from 0.061 in Column (1) to 0.118 in Column (3), which reflects a higher level of homogeneity of firms with the same rating. The magnitude of pseudo R^2 is consistent with that of Kronlund (2020). The coefficient on the high-demand period (γ'_{HD}) is positive and statistically significant, suggesting BBB+ firms are more likely to be upgraded in the high-demand period. Overall, the results are consistent with the notion that investors' demand for A- securities driven by rating-based capital requirements results in abnormally more upgrades of BBB+ firms to the A- category.

To understand the economic significance of the abnormal upgrades induced by rating-based capital requirements, we also estimate the probabilities of upgrades for BBB+ firms and firms with other ratings. The results are reported in Panel B of Table 3. For the full sample, the difference in the probabilities of upgrades for BBB+ firms and other firms is 1.82% in the 2012–2013 period, but -5.86% in other periods (i. e., a difference of 7.68%). For the sample of BBB+ firms and firms with close ratings, the results are very similar. For the BBB+ sample, the difference in the probabilities of upgrades is 14.63% in the 2012–2013 period, but only 5.52% in other periods (i. e., a difference of 9.11%). Untabulated statistics show that the average probability of upgrades for firms is 11.17% during the 2003–2016 period, and 7.02% for firms rated BBB+ at the end of the previous year. The results indicate that the impact of regulation-induced demand for A- bonds on rating outcomes is significant in economic magnitude.

2.2.2. Annual upgrades

To better examine the pattern in rating upgrades when the regulatory advantage of A- securities was high, we further estimate the yearly probabilities of upgrades. This switch to an annual focus facilitates the inclusion and interpretation of year indicators and allows for a more straightforward comparison of upgrade probabilities across years. The regression models for the full sample, the sample of BBB+ firms and firms with ratings close to BBB+, and the sample of BBB+ firms are specified as follows.

$$Prob(UP_{it} = 1) = \psi\left(\alpha + \beta X_{it} + \tau BBB_{+it} + \sum \gamma_t Year_t + \sum \theta_t Year_t \times BBB_{+it} + \varepsilon_{it}\right) \quad (3)$$

$$Prob(UP_{it} = 1) = \psi\left(\alpha' + \beta' X_{it} + \tau' BBB_{+it} + \sum \gamma'_t Year_t + \varepsilon'_{it}\right) \quad (4)$$

In Model (3), the sum of τ and θ_t measure the differences between the

¹⁵ To ensure our findings are not unique to S&P, we also examine the likelihood for Moody's to upgrade bond ratings. The results of untabulated tests show that Moody's upgraded abnormally more Baa1 bonds to A3 category (i. e., equivalent to BBB+ to A- under S&P's rating scale) during the 2013–2014 period, with most of these upgrades occurring in the first half of the 2014, especially in January. These findings confirm that our results are not driven by a specific S&P policy but reflect rating agencies' broader responses to the demand shock for A- securities.

likelihood of a BBB+ firm being upgraded and that of other firms in year t . The difference between θ_h and θ_k captures the difference in the likelihood of being upgraded for BBB+ firms in year h and year k ($h \neq k$). In Model (4), the difference between γ'_h and γ'_k ($h \neq k$) captures the difference in the likelihood of being upgraded for BBB+ firms in year h and year k .

The regression results are reported in Panel A of Table 4. Columns (1) and (2) report the results for the full sample and the sample of firms with ratings at or near BBB+ (i.e., A-, BBB+, BBB, and BBB-). Column (3) reports the results for the sample of BBB+ firms. The standard errors are clustered at the firm level. We set the year 2013 as the base year, so the coefficients on $Year \times BBB+$ in Model (3) and those on $Year$ in Model (4) are the ones of interest.

All the coefficients on the interaction terms of the indicator variable for year t ($t \neq 2013$) and BBB+ are negative and significant in Columns (1), except that for the year 2005.¹⁶ The results in Column (2) are similar, except that the coefficient on θ_{2012} is marginally significant (p -value=0.113) and that on θ_{2015} is only significant based on the one-sided tests (p -value=0.160).¹⁷ In Column (3), all the coefficients on year t ($t \neq 2013$) are negative and significant, except that for the year 2010 the difference is only significant based on one-sided test (p -value=0.138). The pseudo- R^2 rises from 0.070 in Column (1) to 0.157 in Column (3), reflecting the higher homogenous nature of the credit risk of the sample of firms used for the third regression.

Panel B of Table 4 reports the estimated probabilities of upgrades for BBB+ firms and other firms. Based on the results of the regression using the full sample, the upgrade probability for BBB+ firms reaches the lowest point of 1.04% in 2004, while the highest probability of 22.65% occurred in 2013. The probabilities of upgrades for other firms are much more stable and are higher than that for BBB+ firms in most of the years. The difference between the probabilities of upgrades for BBB+ firms and other firms is 7.84% in 2013, which is the highest across years and much larger than those in other years. The estimated results are similar for the sample of firms with ratings close to BBB+. As for the regression using the sample of BBB+ firms, the probability of upgrades for the year 2013 is 20.89%, which is much higher than the estimated probabilities of upgrades for all other years. The results consistently show that compared to the upgrades for BBB+ firms in other years, the upgrades in 2013 were abnormally high.

2.2.3. Upgrades across other ratings

It is possible that the abnormal upgrades of BBB+ firms are caused by some macro-level shocks that also affect firms with other ratings.¹⁸ To avoid this concern, we perform the upgrade tests using samples of firms with other ratings in the previous year, including A-, BBB, BBB-, BB+,

¹⁶ This can be explained by the fact that the demand for US Treasuries was also high, albeit to a less degree, in 2005. Previous literature has established that the inflow of funds from foreign central banks and private investors led to significant increases in demand for US Treasuries around 2005. Warnock and Warnock (2009) estimate that the surge in the inflow of foreign capital depressed the yield of 10-year US Treasury by about 92 basis points. Meanwhile, Kaminska and Zinna (2020) show that the cumulative effect of foreign capital inflow on 10-year Treasury yield up to 2008 was smaller than the effect of the Fed's QE and MEP programs in 2008-2012. The former effect is estimated to be a yield drop of 80 basis points, while the latter 140 basis points, indicating a far stronger incentive to reach-for-yield in the latter period. These findings are largely consistent with our results that abnormal upgrades are more likely to occur when excessive demand for Treasury securities leads to more demand for A- bonds.

¹⁷ The spread between A- and BBB+ firms was relatively high in 2015, though lower than that in 2012 and 2013. But as the prevailing interest in the market rose, the overall regulatory advantage of A- firms in 2015 was lower than that in 2013.

¹⁸ For example, lowered interest rates benefit many firms and thus CRAs tend to upgrade more firms across ratings in 2013 than in other years.

BB, BB-, B+, B, and B-.¹⁹ The results are reported in Table 5. Recall that we set 2013 as the base year, so the coefficients on the year indicators represent the differences between firms' likelihood of being upgraded in 2013 and those in other years. As reported, the upgrades of firms with ratings other than BBB+ do not exhibit the same abnormal pattern as we find for the sample of BBB+ firms. Most of the coefficients on the year indicators are not statistically significant. Overall, these results do not support the alternative hypothesis that the abnormal upgrades of BBB+ firms in 2013 are driven by some unknown shocks common to other ratings as well.

Moreover, this set of tests also helps exclude the possibility that our results are driven by S&P's review of rating criteria in 2013. In November 2013, S&P reviewed and updated its rating methodology for corporate entities (Standard and Poor's Rating Services, 2013). The primary purpose of the review is to enhance the transparency and comparability of ratings. From this update in methodology, S&P expected to change only a very small proportion of the existing ratings.²⁰ Bonsall (2014) indicate that rating methodology could affect rating properties. If the systematic review of credit rating methodology has significant effects on ratings, we expect to find abnormal upgrades across different credit rating categories in 2013 relative to other years. That means the abnormal upgrade pattern we find for BBB+ firms should also be observed for firms with other ratings.

But the results lend no support to the notion that the effect of the review by S&P on credit ratings is systematic across different credit rating categories. The methodology review conducted by the S&P cannot explain the abnormal upgrades of BBB+ firms in 2013. Based on the evidence, we cannot exclude the possibility that S&P takes advantage of the updates on rating criteria to issue favorable ratings to some BBB+ firms. This possibility is certainly consistent with our contention of rating inflation around the A- cutoff in 2013, and it is actually similar to the findings of Beatty et al. (2019) that rating agencies use their discretion over rating calibration to achieve their own objective. According to their study, the clients of Moody's and Fitch receive better ratings after the two rating agencies recalibrate the ratings of municipal bonds.

2.2.4. Expected rating scores

To further examine the pattern of the upgrades of BBB+ firms in 2013, we look at their expected rating scores. To calculate the expected rating scores, we follow Baghai et al. (2014) and regress firms' actual rating scores on the same set of control variables as in the previous tests as well as the firm and year indicators. The actual rating scores are integer numbers assigned to each rating. Specifically, AAA+ is coded 1, and C is coded 21. Other ratings are coded accordingly. The expected rating scores are the values predicted by the model. We further estimate the following model using the sample of all firms rated BBB+ (A-) at the end of the previous (current) year with firm fixed effects.

$$RatExp_{it} = \alpha_{Exp} + \beta_{Exp} UP_{it} + \tau_{Exp} Year2013_{it} + \gamma_{Exp} UP_{it} \times Year2013_{it} + \sum \lambda_{Exp,i} + \varepsilon_{Exp,it} \quad (5)$$

¹⁹ We do not perform the tests on samples of firms with other ratings as the sample sizes are very small.

²⁰ According to S&P, the purpose of the methodology update was to "organize the analytical process according to a common framework and articulate the steps in developing the stand-alone credit profile (SACP) and issuer credit rating (ICR) for a corporate entity." S&P also estimated that "approximately 10% of corporate industrial companies and utilities ratings within the scope of the criteria to change. Of that number, we expect between 80%-90% to receive a one-notch change, with the majority of the remainder receiving a two-notch change. The number of upgrades would approximately equal the number of downgrades." For more details, see S&P's request for comment at https://www.standardandpoors.com/en_US/web/guest/article/-/view/type/HTML/i/1219279.

Table 3
Abnormal upgrades during the high-demand period.

Panel A: Regressions						
Variables	(1)		(2)		(3)	
	Full Sample		Close Ratings		BBB±	
	Coef.	z-stat	Coef.	z-stat	Coef.	z-stat
BBB+ × HighDemand	0.958***	4.078	1.026***	4.000		
BBB+	-0.705***	-5.598	-0.625***	-4.683		
HighDemand	0.144*	1.857	0.101	0.672	1.160***	4.787
TrendTd	-0.030	-0.815	-0.327***	-3.746	-0.281	-1.528
TrendDlt	-0.040	-1.166	0.014	0.186	-0.207	-1.336
TrendROA	0.532***	6.762	0.727***	4.510	0.800*	1.885
TrendIntcov	0.058***	3.264	0.007	0.225	-0.006	-0.078
DLeverage	-3.730***	-8.377	-2.546**	-2.259	-2.422	-1.001
DConvert	-0.298	-0.275	1.283	0.405	3.841	0.439
DRent	4.144	1.145	-8.380	-0.634	-41.622*	-1.771
DCash	-0.214	-0.389	-0.941	-0.828	-3.595	-1.310
DDebt	0.085***	3.763	0.075	1.091	0.124	1.066
DIntCov	-1.228***	-4.515	-0.511	-0.547	-2.392	-1.356
DProfit	0.780*	1.709	1.757**	2.013	1.887	0.763
DSize	1.025***	6.627	0.189	0.641	0.276	0.420
DPPE	-0.561	-0.773	-1.496	-0.893	-0.273	-0.043
DCapital	4.268***	3.514	5.495**	2.271	5.037	0.832
DProVol	1.892	1.420	4.604	1.488	-2.449	-0.388
Leverage	-0.077	-0.476	-0.619	-1.108	-1.688	-1.093
Convert	0.069	0.116	0.855	0.433	6.764	1.425
Rent	1.035	0.977	-1.743	-0.501	-11.897	-0.873
Cash	0.991***	3.063	1.152*	1.858	2.391	1.517
Debt	-0.153***	-5.589	-0.024	-0.294	0.130	0.960
IntCov	1.578***	4.889	-2.217*	-1.772	-7.999***	-3.105
Profit	0.761***	3.252	1.656***	4.158	1.564	1.362
Size	-0.052**	-2.220	0.181***	3.981	0.261**	2.500
PPE	-0.375**	-2.336	0.038	0.141	1.504**	2.227
Capital	0.628	0.807	-0.181	-0.112	-4.902	-1.234
ProVol	0.176	0.423	-4.287***	-2.621	-5.865*	-1.683
Predown	0.571***	7.606	0.169	1.272	-0.051	-0.162
Constant	-1.830***	-8.161	-3.893***	-8.540	-5.033***	-4.617
Observations	15,185		5923		1424	
Pseudo R ²	0.061		0.065		0.118	

Panel B: Estimated probabilities of upgrades				
	Periods	BBB+ Firms	Other Firms	Diff.
Full Sample	High Demand	14.67%	12.85%	1.82%
	Low Demand	5.53%	11.39%	-5.86%
Close Ratings Sample	High Demand	14.80%	9.65%	5.15%
	Low Demand	5.51%	8.84%	-3.33%
BBB+ Sample	High Demand	14.63%		
	Low Demand	5.52%		

Panel A of this table reports the results of logistic regressions using the full sample, the sample of firms rated A-, BBB+, BBB or BBB- in the previous year, and the sample of firms rated BBB+ in the previous year, respectively. The dependent variable is an indicator variable equal to one if a firm is upgraded in the current year, and zero otherwise. See [Appendix C](#) for definitions of variables. Panel B reports the estimated probabilities of upgrades. Standard errors are clustered at the firm level. ***, ** and * denote statistical significance at the 1%, 5% and 10% levels, respectively.

RatExp is the expected rating scores. Panel A of [Table 6](#) reports the estimation results. The results in Columns (1) and (2) are from the estimations using the BBB+ and A- samples mentioned above, respectively. Firms that were downgraded during 2007–2009 were excluded from the samples. We find that the coefficient on $UP \times Year2013$ is positive but statistically insignificant for the regression estimated using the sample of BBB+ firms, while it is significantly positive for that estimated using the sample of A- firms. The results indicate that the expected ratings of the BBB+ firms upgraded in 2013 (i.e., A- in 2013) are lower than those of other firms rated A- in 2013 (i.e., the upgraded BBB+ firms have higher expected rating scores), while such a pattern does not exist if we compare the expected ratings of the upgraded BBB+ firms with those of other firms rated BBB+ in 2012.

We also test the differences in the expected rating scores between the upgraded BBB+ firms and other BBB+ (A-) firms by year. Panel B of [Table 6](#) reports the differences in expected rating scores between the upgraded BBB+ firms and other BBB+ (A-) firms (i.e., non-upgraded minus upgraded) and the *t*-test results. A higher expected rating score

suggests lower creditworthiness. As shown in Panel B, the expected rating scores of the non-upgraded BBB+ firms are significantly higher than those of the upgraded BBB+ firms for many years (i.e., lower creditworthiness), but not for 2013. In the sample of firms rated A- in the current year, for years other than 2013, the expected rating scores of the upgraded BBB+ firms are either not statistically different from or significantly lower than those of other A- firms (i.e., equal or higher creditworthiness). But for 2013, the expected rating scores of the upgraded BBB+ firms are significantly higher than those of other A- firms (i.e., lower creditworthiness). Overall, the results are consistent with the notion that the upgrades of BBB+ firms in 2013 are different from those in other years.

3. Financial performance and policies of the upgraded BBB+ firms

We further examine five financial indicators that are essential to credit rating assignment around the upgrades of the BBB+ firms in 2013:

Table 4
Abnormal upgrades in 2013.

Panel A: Regressions									
Variables	(1)			(2)			(3)		
	Full Sample			Close Ratings			BBB±		
	Coef.	z-stat	p-value	Coef.	z-stat	p-value	Coef.	z-stat	p-value
BBB+ × Year 2003	-1.406**	-2.154	0.031	-1.340*	-1.903	0.057			
BBB+ × Year 2004	-2.618**	-2.470	0.014	-2.489**	-2.270	0.023			
BBB+ × Year 2005	-0.482	-1.130	0.258	-0.806*	-1.706	0.088			
BBB+ × Year 2006	-1.587***	-2.717	0.007	-1.703***	-2.699	0.007			
BBB+ × Year 2007	-2.031***	-3.132	0.002	-2.338***	-3.412	0.001			
BBB+ × Year 2008	-1.258**	-2.549	0.011	-0.956*	-1.696	0.090			
BBB+ × Year 2009	-1.808**	-2.333	0.020	-2.068**	-2.537	0.011			
BBB+ × Year 2010	-0.788*	-1.830	0.067	-0.795*	-1.691	0.091			
BBB+ × Year 2011	-1.443***	-2.826	0.005	-1.659***	-2.890	0.004			
BBB+ × Year 2012	-0.782*	-1.793	0.073	-0.794	-1.583	0.113			
BBB+ × Year 2014	-1.540***	-2.860	0.004	-1.346**	-2.340	0.019			
BBB+ × Year 2015	-0.773*	-1.736	0.083	-0.709	-1.406	0.160			
BBB+ × Year 2016	-1.215**	-2.410	0.016	-1.542***	-2.884	0.004			
Year 2003	-0.745***	-4.789		-0.508	-1.591		-1.926**	-2.506	0.012
Year 2004	-0.836***	-5.430		-1.006***	-3.241		-3.719***	-3.362	0.001
Year 2005	-0.453***	-3.155		-0.237	-0.865		-1.284***	-2.662	0.008
Year 2006	-0.459***	-3.142		-0.354	-1.226		-2.398***	-3.783	0.000
Year 2007	-0.177	-1.254		0.096	0.350		-2.546***	-3.527	0.000
Year 2008	-0.142	-0.972		-0.416	-1.357		-1.405***	-2.842	0.005
Year 2009	-0.576***	-3.598		-0.234	-0.771		-2.265***	-2.801	0.005
Year 2010	-0.000	-0.002		0.041	0.156		-0.651	-1.485	0.138
Year 2011	-0.133	-1.000		-0.043	-0.166		-1.802***	-3.283	0.001
Year 2012	-0.274**	-2.015		-0.235	-0.861		-1.014**	-2.339	0.019
Year 2014	-0.104	-0.765		-0.365	-1.360		-1.757***	-3.242	0.001
Year 2015	-0.287*	-1.945		-0.455	-1.541		-1.115**	-2.567	0.010
Year 2016	-0.285*	-1.928		0.043	0.165		-1.465***	-2.948	0.003
BBB+	0.555**	2.181		0.716**	2.390				
Constant	-1.364***	-5.486		-3.538***	-6.916		-3.208***	-2.666	
Observations	15,185			5923			1424		
Controls	Yes			Yes			Yes		
Pseudo R ²	0.070			0.079			0.157		

Panel B: Estimated probabilities of upgrades							
Year	Full Sample			Close Ratings			BBB±
	BBB+ Firms	Other Firms	Diff.	BBB+ Firms	Other Firms	Diff.	BBB+ Firms
2003	3.62%	7.88%	-4.26%	4.04%	8.52%	-4.48%	4.42%
2004	1.04%	7.27%	-6.23%	0.79%	5.51%	-4.72%	0.80%
2005	10.82%	10.16%	0.66%	7.88%	11.66%	-3.78%	7.76%
2006	4.00%	10.11%	-6.11%	2.99%	8.54%	-5.55%	2.86%
2007	3.43%	12.82%	-9.38%	2.45%	12.01%	-9.56%	2.49%
2008	7.21%	13.20%	-5.98%	5.77%	8.82%	-3.05%	7.00%
2009	2.91%	9.14%	-6.23%	2.41%	10.81%	-8.41%	3.23%
2010	12.24%	14.81%	-2.57%	10.48%	13.67%	-3.19%	12.98%
2011	6.16%	13.30%	-7.14%	4.36%	12.73%	-8.37%	4.94%
2012	9.75%	11.82%	-2.07%	8.06%	10.64%	-2.58%	9.72%
2013	22.65%	14.81%	7.84%	18.61%	11.47%	7.14%	20.89%
2014	5.79%	13.62%	-7.82%	4.37%	9.96%	-5.60%	5.14%
2015	9.72%	11.70%	-1.98%	7.11%	7.78%	-0.67%	8.94%
2016	6.59%	11.71%	-5.12%	5.28%	13.60%	-8.32%	6.64%

Panel A of this table reports the results of logistic regressions using the full sample, the sample of firms rated A-, BBB+, BBB or BBB- in the previous year, and the sample of firms rated BBB+ in the previous year. The dependent variable is an indicator variable equal to one if a firm is upgraded in the current year, and zero otherwise. See Appendix C for definitions of variables. Panel B reports the estimated probabilities of upgrades. Standard errors are clustered at the firm level. ***, ** and * denote statistical significance at the 1%, 5% and 10% levels, respectively.

ROA, profit margin (*PMargin*), leverage (*Leverage*), the reverse of interest coverage ratio (*IntCov*), and cash flow divided by total debt (*CFODebt*). ROA and *PMargin* measure firms' financial performance; *Leverage*, *IntCov*, and *CFODebt* measure firms' ability to pay off debt. To assess the performance of BBB+ firms that were upgraded in 2013, we use the BBB+ firms without rating changes in 2013 as the control group. The regression model is specified below.

$$\text{Measure}_{it} = \alpha_{Fi} + \gamma_{Fi} \text{UP2013}_{it} \times \text{Post}_{it} + \theta_{Fi} S_{izE_{it}} + \sum \lambda_{Fi_i} + \sum \eta_{Fi_j,t} + \varepsilon_{Fi_{it}} \quad (6)$$

The coefficient of interest is the one on $\text{UP2013} \times \text{Post}$. *UP2013* is an indicator variable equal to one if a BBB+ firm was upgraded in 2013, and zero otherwise. *Post* is an indicator variable equal to one for the period of 2013–2015, and zero otherwise. If the upgrades reflect improvement in the performance measures, the coefficient on $\text{UP2013} \times \text{Post}$ should be positive (negative) and significant for ROA, *PMargin*, and *CFODebt* (*Leverage* and *IntCov*). We also control for firm size, and firm and year fixed effects. The sample includes all the firms that were rated BBB+ at the end of year 2012 for 2011–2015 and contains 578 firm-year observations. We cluster the standard errors at the firm level.

The Columns (1)–(5) of Table 7 show that none of the coefficients on $\text{UP2013} \times \text{Post}$ are statistically significant across the regressions,

Table 5

Upgrades of firms with other ratings.

Variables	(1)		(2)		(3)		(4)		(5)	
	<u>A-</u>		<u>BBB</u>		<u>BBB-</u>		<u>BB±</u>		<u>BB</u>	
	Coef.	z-stat	Coef.	z-stat	Coef.	z-stat	Coef.	z-stat	Coef.	z-stat
Year 2003	-0.129	-0.129	-0.816	-1.400	0.011	0.024	-0.393	-0.696	-0.227	-0.433
Year 2004	-0.995	-1.157	-0.878*	-1.686	-0.641	-1.341	-0.863	-1.619	-0.736	-1.291
Year 2005	-0.588	-0.717	0.232	0.529	-0.104	-0.250	-0.104	-0.216	-1.000*	-1.755
Year 2006	0.540	0.790	-0.096	-0.199	-0.490	-1.042	-0.471	-0.905	-0.603	-1.197
Year 2007	0.988	1.511	0.448	1.027	-0.344	-0.742	-0.300	-0.526	-0.268	-0.515
Year 2008	-0.480	-0.558	-0.339	-0.666	-0.368	-0.829	0.603	1.185	0.493	1.047
Year 2009	-0.694	-0.764	-0.339	-0.631	0.074	0.164	-0.679	-1.059	-1.502**	-2.133
Year 2010	-0.970	-1.119	0.598	1.433	-0.086	-0.198	-0.296	-0.568	0.069	0.150
Year 2011	-1.263	-1.499	0.320	0.745	-0.036	-0.092	-1.056*	-1.854	0.711*	1.701
Year 2012	-1.195	-1.272	-0.122	-0.274	-0.063	-0.155	-0.658	-1.444	0.127	0.286
Year 2014	-1.153	-1.453	-0.061	-0.141	-0.269	-0.625	-1.819**	-2.556	0.509	1.178
Year 2015	-0.534	-0.759	-0.722	-1.324	-0.282	-0.623	-0.867*	-1.700	0.441	0.989
Year 2016	-0.745	-1.041	0.745*	1.809	-0.502	-1.099	-1.161**	-2.278	-0.410	-0.754
Observations	981		2084		1434		948		1346	
Controls	Yes		Yes		Yes		Yes		Yes	
Pseudo R ²	0.172		0.110		0.083		0.164		0.149	

Variables	(6)		(7)		(8)		(9)	
	<u>BB-</u>		<u>B±</u>		<u>B</u>		<u>B-</u>	
	Coef.	z-stat	Coef.	z-stat	Coef.	z-stat	Coef.	z-stat
Year 2003	-0.463	-1.100	-0.351	-0.844	-0.174	-0.312	-1.548	-1.546
Year 2004	-1.077**	-2.467	-0.693*	-1.681	0.104	0.222	-0.702	-0.807
Year 2005	-0.192	-0.522	-0.814*	-1.897	0.228	0.479	-0.521	-0.657
Year 2006	-0.650	-1.520	-0.473	-1.294	0.298	0.656	-0.281	-0.343
Year 2007	-0.836**	-1.993	-0.284	-0.770	0.174	0.390	0.061	0.081
Year 2008	-0.162	-0.406	-0.480	-1.179	-0.242	-0.475	-0.066	-0.081
Year 2009	-0.639	-1.521	-1.655***	-3.015	0.101	0.214	-0.046	-0.061
Year 2010	-0.531	-1.291	-0.065	-0.173	0.201	0.532	1.089	1.461
Year 2011	0.079	0.198	-0.145	-0.411	-0.406	-1.038	0.362	0.428
Year 2012	-0.530	-1.334	-0.696*	-1.681	0.028	0.072	0.268	0.345
Year 2014	0.029	0.074	0.099	0.260	0.206	0.543	-0.200	-0.276
Year 2015	-0.771*	-1.695	-0.562	-1.313	0.539	1.372	-0.340	-0.422
Year 2016	-0.590	-1.303	-0.161	-0.401	0.148	0.316	-0.425	-0.530
Observations	1654		1506		1229		561	
Controls	Yes		Yes		Yes		Yes	
Pseudo R ²	0.133		0.168		0.135		0.240	

This table reports the results of logistic regressions using the samples of firms rated A- through B- (except for BBB+) in the previous year. The dependent variable is an indicator variable equal to one if a firm is upgraded in the current year, and zero otherwise. See [Appendix C](#) for definitions of variables. Standard errors are clustered at the firm level. ***, ** and * denote statistical significance at the 1%, 5% and 10% levels, respectively.

suggesting that there is no evidence that the financial performance and ability to pay off debt of the upgraded BBB+ firms improved relative to other BBB+ firms around the upgrades. We admit that rating upgrades could lag issuers' financial improvement, but our results still hold if we also code *Post* to be one for the year 2012.

We also examine how firms' interest expenses divided by total debt (*IntExp*) change after the upgrades. The regression specification is the same as Model (6). As Column (6) in [Table 6](#) indicates, *IntExp* decreases for the upgraded BBB+ firms relative to other BBB+ firms, which is consistent with that the upgrades help the upgraded firms to lower their interest burden.

We further look at the changes in the investments and total payouts of the upgraded BBB+ firms. Firm investments are defined as the sum of capital expenditure and M&A expenses. Total payouts are the sum of cash dividends and net share repurchases. As the results of Columns (7) and (8) show, the investments remained stable for the upgraded BBB+ firms, but their total payouts increased after the upgrades. Overall, these results are consistent with that the upgraded firms take advantage of the upgrades to lower their interest burden and increase payouts to investors rather than expand investments.

4. Interest burdens and upgrades

If the upgrades of BBB+ firms in 2013 were driven by the low-interest environment, we predict that firms with high interest burdens

will be more likely to be upgraded because they will benefit more from rating upgrades. To test this prediction, we measure firms' interest burden using three variables: interest expenses divided by total debt (*Int/TDebt*), the reverse of interest coverage ratio (*IntCov*), and total debt divided by EBITDA (*Debt*). We then estimate the following logistic model using the sample of firms that are rated BBB+ at the end of the previous year. *Measure* is one of the three measures of firms' interest burden. *High_Measure* is an indicator variable equal to one if the interest expense measure for a firm is higher than the median level for the year and for the same rating category, and zero otherwise.

$$Prob(UP_{it}=1) = \psi_{IN}(\alpha_{IN} + \beta_{IN}High_Measure_{it} + \tau_{IN}Year2013_{it} + \gamma_{IN}High_Measure_{it} \times Year2013_{it} + \theta_{IN}Controls_{it} + \varepsilon_{IN,it}) \quad (7)$$

The results are reported in [Table 8](#). The coefficients on the interactions of the high interest burden measures and the indicator of the year 2013 are all significantly positive across the three regressions, suggesting that BBB+ firms with higher interest burdens were more likely to be upgraded in 2013, which is consistent with our prediction.

An alternative explanation is that the firms with higher interest burdens benefit more from low interest costs during the quantitative easing programs, so they are more likely to be upgraded. If this explains our findings, we should find abnormal upgrades across many rating categories. But we only find abnormal upgrades among BBB+ firms.

Table 6
Expected rating scores.

Panel A: Regressions on expected rating scores								
Variables	(1)				(2)			
	<u>BBB±</u>				<u>A-</u>			
	Coef.	t-stat			Coef.	t-stat		
<i>UP</i> × <i>Year2013</i>	0.081			1.006	0.109*			1.815
<i>UP</i>	-0.217***			-4.405	-0.041			-0.900
<i>Year2013</i>	0.024			0.732	0.016			0.866
Observations	1244				919			
No. of Firms	335				252			
Firm FE	Yes				Yes			
R ² (Within)	0.027				0.001			

Panel B: Differences in expected rating scores between the non-upgraded and upgraded firms								
Year	BBB+				A-			
	Dif.	t-stat	P(Dif.<0)	P(Dif.>0)	Dif.	t-stat	P(Dif.<0)	P(Dif.>0)
2003	1.025	1.462	0.926	0.074	0.838	1.116	0.866	0.134
2004	NA	NA	NA	NA	NA	NA	NA	NA
2005	1.491	3.514	1.000	0.000	1.111	2.759	0.996	0.004
2006	0.574	0.859	0.804	0.196	0.323	0.433	0.667	0.333
2007	0.303	0.421	0.663	0.337	-0.078	-0.089	0.465	0.535
2008	0.859	1.809	0.963	0.037	0.659	1.157	0.874	0.126
2009	1.116	1.430	0.922	0.078	0.926	0.983	0.835	0.165
2010	0.335	0.823	0.793	0.207	-0.207	-0.407	0.343	0.657
2011	0.969	1.647	0.948	0.052	0.546	0.737	0.768	0.232
2012	0.309	0.734	0.768	0.232	-0.110	-0.213	0.416	0.584
2013	0.042	0.141	0.556	0.444	-0.374	-1.316	0.096	0.904
2014	1.156	1.810	0.963	0.037	0.900	1.440	0.923	0.077
2015	0.701	1.441	0.923	0.077	0.189	0.436	0.668	0.332
2016	0.158	0.208	0.582	0.418	-0.178	-0.293	0.385	0.615

Panel A of this table reports the OLS regressions of estimated rating scores on the indicators for upgraded firms and year 2013 as well as their interaction using the sample of firms rated BBB+ (A-) in the previous (current) year. Firms that were downgraded during 2007–2009 are excluded. Panel B reports the results of *t*-tests that compare the expected rating scores of upgraded and other BBB+ (A-) firms for each year (Dif.=non-upgraded-upgraded). See [Appendix C](#) for definitions of variables. Standard errors are clustered at the firm level. ***, ** and * denote statistical significance at the 1%, 5% and 10% levels, respectively.

Table 7
Changes in financial performance and financial policies.

Variables	(1)		(2)		(3)		(4)	
	<u>ROA</u>		<u>ProfitMargin</u>		<u>Leverage</u>		<u>IntCov</u>	
	Coef.	t-stat	Coef.	t-stat	Coef.	t-stat	Coef.	t-stat
<i>UP2013</i> × <i>Post</i>	0.003	0.592	-0.003	-0.265	-0.010	-0.982	-0.004	-0.358
<i>Size</i>	-0.030*	-1.818	0.066*	1.845	0.038	0.758	-0.004	-0.053
<i>Constant</i>	0.409***	2.617	-0.382	-1.121	-0.078	-0.164	0.178	0.241
Observations	578		578		578		578	
No. of Firms	123		123		123		123	
Firm FE	Yes		Yes		Yes		Yes	
Year FE	Yes		Yes		Yes		Yes	
R ² (Within)	0.070		0.030		0.132		0.003	

Variables	(5)		(6)		(7)		(8)	
	<u>CFO/Debt</u>		<u>IntExp</u>		<u>Investments</u>		<u>Payouts</u>	
	Coef.	t-stat	Coef.	t-stat	Coef.	t-stat	Coef.	t-stat
<i>UP2013</i> × <i>Post</i>	0.003	0.026	-0.004*	-1.973	-0.003	-0.706	0.013*	1.884
<i>Size</i>	-0.376**	-2.544	-0.011	-1.593	0.009	0.542	-0.033*	-1.872
<i>Constant</i>	4.055***	2.874	0.153**	2.379	-0.030	-0.190	0.360**	2.149
Observations	578		578		578		578	
No. of Firms	123		123		123		123	
Firm FE	Yes		Yes		Yes		Yes	
Year FE	Yes		Yes		Yes		Yes	
R ² (Within)	0.059		0.148		0.009		0.049	

This table reports the results of OLS regressions that examine firm performance and financial policies before and after the upgrades in 2013 for 2011–2015. *UP2013* is an indicator variable equal to one if a BBB+ firm was upgraded in 2013, and zero otherwise. *Post* is an indicator variable equal to one for the period 2013–2015, and zero otherwise. *Size* is the logarithm of the book value of assets. Standard errors are clustered at the firm level. ***, ** and * denote statistical significance at the 1%, 5% and 10% levels, respectively.

Table 8
Interest burden and likelihood of upgrade.

Variables	(1)		(2)		(3)	
	Coef.	z-stat	Coef.	z-stat	Coef.	z-stat
<i>High_Int/TDebt</i> × <i>Year 2013</i>	1.242**	2.078				
<i>High_IntCov</i> × <i>Year 2013</i>			1.719***	2.875		
<i>High_Debt</i> × <i>Year 2013</i>					1.908***	3.180
<i>High_Int/TDebt</i>	-0.013	-0.044				
<i>High_IntCov</i>			-0.279	-0.816		
<i>High_Debt</i>					-0.528*	-1.738
<i>Year 2013</i>	0.747	1.564	0.599	1.364	0.539	1.248
Observations	1423		1424		1424	
Controls	Yes		Yes		Yes	
Pseudo R ²	0.131		0.136		0.139	

This table reports the results of tests that examine the impact of firms' interest burden on their likelihood of being upgraded. The dependent variable is *UP*. *High_Int/TDebt* (*High_IntCov*, *High_Debt*) is an indicator variable equal to one if a firm's interest expense divided by total debt (*IntCov*, *Debt*) is higher than the median level for observations in the same year and rating category, and zero otherwise. *Year 2013* is an indicator variable for year 2013. Control variables are the same with those in the main regressions. See [Appendix C](#) for variable definitions. Standard errors are clustered at the firm level. ***, ** and * denote statistical significance at the 1%, 5% and 10% levels, respectively.

Additionally, the BBB+ firms with higher interest burdens will benefit from the sharp decline in borrowing costs only after they are actually upgraded to the A- category (i.e., after the S&P has made the upgrade decisions). As such, it is unlikely that our results are solely driven by CRAs recognizing the decreased default risks of BBB+ firms with higher interest burdens.

5. CRAs' incentive

In this section, we examine the rating agency's incentive to issue

abnormally more upgrades. As prior studies indicate, CRAs often cater to investors to maintain sources of revenues ([Jiang et al., 2012](#); [Cornaggia and Cornaggia, 2013](#)). Following this reasoning, we predict that firms that issue more bonds are able to exert greater pressure on rating agencies, because they are more active clients and are more likely to issue bonds in the future. But if the active clients have issued too much debt, they tend to have lower debt capacity and are likely to have a lower demand for external financing. Therefore, we expect firms that were active clients of S&P and were likely to still have debt capacity were more likely to be upgraded in 2013.

Table 9
Rating agency's incentive.

Panel A: Prior bond issuance and likelihood of upgrades						
Variables	(1)		(2)		(3)	
	<u>UP_Abnormal</u>					
	Coef.	z-stat	Coef.	z-stat	Coef.	z-stat
<i>Active_Client</i>	1.442**	2.286	1.229 ⁺	1.569	0.327	0.653
<i>Active_Client</i> × <i>Year 2013</i>					1.111 ⁺	1.342
<i>Year 2013</i>					1.563***	4.273
Observations	126		121		1509	
Controls	No		Yes		Yes	
Pseudo R ²	0.045		0.215		0.106	
Variables	(4)		(5)		(6)	
	<u>UP_Abnormal2</u>					
	Coef.	z-stat	Coef.	z-stat	Coef.	z-stat
<i>Active_Client</i>	1.266*	1.693	1.430 ⁺	1.446	0.040	0.051
<i>Active_Client</i> × <i>Year 2013</i>					1.506 ⁺	1.320
<i>Year 2013</i>					1.817***	3.610
Observations	126		121		1509	
Controls	No		Yes		Yes	
Pseudo R ²	0.033	0.239	0.209			
PPanel B: Abnormal upgrades and future bond issuance						
Variables	(1)		(2)		(3)	
	Coef.	t-stat	Coef.	t-stat	Coef.	t-stat
<i>AbUP_Client</i>	6.040***	2.856	6.577***	3.037		
<i>AbUP_Client2</i>					6.744**	2.471
Observations	126		126		126	
Controls	No		Yes		No	
Adjusted R ²	0.054		0.117		0.039	

Panel A of this table reports the results of tests on the relationship between previous bond issuance and the likelihood of abnormal upgrades. The dependent variables are indicator variables equal to one if a BBB+ firm was upgraded to A- but the estimated rating score is greater than the actual rating score by one notch or two notches (*UP_Abnormal* or *UP_Abnormal2*). *Active_Client* is an indicator variable equal to one if a firm issued bonds in the three years prior to the upgrades but the amounts were less than the median level. Panel B reports the results of tests on the relationship between abnormal upgrades and future bond issuance. The dependent variable is the logarithm of bond issuance in 2014. *AbUP_Client* equals to one if a previously identified borrower with debt capacity was abnormally upgraded in 2013. See [Appendix C](#) for definitions of variables. Standard errors are clustered at the firm level. ***, ** and * denote statistical significance at the 1%, 5% and 10% levels, respectively.

Table 10
Comparison with Egan-Jones ratings.

Panel A: Observations with Egan-Jones Ratings								
Yes		Observations						
2013	403							
2014	423							
2015	390							
2016	366							
Total	1583							
Panel B: Regressions								
	(1)		(2)		(3)		(4)	
Variables	Coef.	t-stat	Coef.	t-stat	Coef.	t-stat	Coef.	t-stat
$A \times \textit{Year 2013}$	0.346**	2.088						
$A- \times \textit{Year 2013}$			-0.243*	-1.722				
$BBB+ \times \textit{Year 2013}$					-0.024	-0.170		
$BBB \times \textit{Year 2013}$							-0.195	-1.455
A	-0.055	-0.283						
A-			-0.363**	-2.142				
BBB+					-0.016	-0.096		
BBB							0.097	0.705
<i>Year 2013</i>	-0.053	-0.836	0.013	0.208	-0.010	-0.150	0.025	0.386
Constant	7.145**	2.380	7.706***	2.611	7.155**	2.373	7.208**	2.405
Observations	1582		1582		1582		1582	
No. of Firms	459		459		459		459	
Controls	Yes		Yes		Yes		Yes	
Firm FE	Yes		Yes		Yes		Yes	
R ² (Within)	0.291		0.296		0.288		0.290	

Panel A of this table reports the yearly distribution of the firm-year observations with both S&P and Egan-Jones credit ratings. Panel B reports the results from regressions of the differences between S&P and Egan-Jones rating scores on indicator variables for A through BBB firms, *Year 2013*, and their interactions. Control variables are the same with those in the main tests. See Appendix C for variable definitions. Standard errors are clustered at the firm level. ***, ** and * denote statistical significance at the 1%, 5% and 10% levels, respectively.

To test this prediction, we identify firms that have issued bonds in the prior three years but the amounts issued are less than the median of the total amounts issued by BBB+ firms (i.e., rated BBB+ at the end of 2012) during this period (*Active_Client*). We consider these firms as being influential to CRAs and still having debt capacity. Since we focus on rating inflation, we also identify inflated ratings as those of which the actual rating score is less than the estimated rating score by one or two notches (denoted as *UP_Abnormal* and *UP_Abnormal2*). The estimation method is the same as described in Section 3.2.4. The regression sample includes all the firms that were rated BBB+ at the end of 2012 (or all the sample years). The regression results are reported in Panel A of Table 9. As the results indicate, the identified CRAs' active clients with debt capacity are significantly more likely to be abnormally upgraded in 2013, which is consistent with our prediction.

We further investigate whether the abnormally upgraded firms (*AbUP_Client* and *AbUP_Client2*) are indeed more likely to issue bonds after being upgraded. Specifically, we examine whether these firms issue larger amounts of bonds in 2014. The sample includes all the firms that were rated BBB+ at the end of 2012. As Panel B of Table 9 presents, BBB+ firms that were abnormally upgraded to the A- category issued larger amounts of bonds in 2014.²¹

6. Comparison with EJR ratings

Our research design exploits the exogenous shock to the long-term interest rate of Treasury securities and thus investors' demand for A-corporate bonds in 2013. While the regressions guided by this design generate evidence in support of our hypothesis, to further consolidate our theory, we compare the ratings of S&P with those of EJR. EJR follows an investor-paid revenue model, which shelters them from pressures from issuers. In addition, risk-adjusted capital requirements

demand that insurance companies use ratings issued by S&P, Moody's or Fitch in the calculation of capital, so EJR is unlikely to face the same pressure as S&P does. Therefore, we expect the ratings of S&P to be higher than those of EJR in 2013.

The empirical test of this prediction requires data on EJR ratings, which is not publicly available before 2012. But the Securities and Exchange Commission (SEC) Regulation 17 g-7 requires that all NRSROs publicly disclose their historical rating assignments in XBRL format since June 15, 2012.²² The Center for Municipal Finance with the support from the Center for Corporate and Securities Law at the University of San Diego School of Law compiles the disclosed historical ratings of all NRSROs, including the ratings issued by EJR. We obtain EJR's ratings from the rating files published by the Center for Municipal Finance.²³ There are only a few EJR ratings available for 2012, so we only keep EJR's ratings for 2013–2016. We do not follow the main research design for the same reason, since we cannot reasonably determine the upgrades issued by EJR in 2013.

Panel A of Table 10 reports the distribution of the firm-year observations that have ratings from both S&P and EJR. We then estimate the following model. *RatDif* is the difference between the rating scores of S&P and EJR for a firm. The scores for the ratings of EJR are coded similarly to those for S&P's ratings. *Rating* is a firm's S&P rating at the end of the year, ranging from A to BBB.

$$\begin{aligned} \text{RatDif}_{it} = & \alpha_{EJ} + \beta_{EJ} \text{Rating}_{it} + \tau_{EJ} \text{Year2013}_{it} + \gamma_{EJ} \text{Rating}_{it} \times \text{Year2013}_{it} \\ & + \theta_{EJ} \text{Controls}_{it} + \sum \lambda_{EJ,i} + \varepsilon_{EJ,it} \end{aligned} \quad (8)$$

Panel B of Table 10 reports the regression results. The coefficient on $A- \times \text{Year } 2013$ is negative and significant (Column 2), suggesting that

²¹ The majority of the bond issuance by these firms occurred in 2014, so we use this simplified research design.

²² For more details on the regulation on publication of rating history files, see <https://www.sec.gov/structureddata/rocr-publication-guide.html>.

²³ The data is available at <http://www.ratingshistory.info>.

Table 11
Robustness tests.

Panel A: Expected Default Forecast						
Variables	(1)		(2)		(3)	
	Coef.	z-stat	Coef.	z-stat	Coef.	z-stat
<i>BBB+ × Year 2003 (Year 2003)</i>	-1.130*	-1.662	-1.120	-1.536	-1.645**	-2.255
<i>BBB+ × Year 2004 (Year 2004)</i>	-2.279**	-2.112	-2.245**	-2.000	-3.724***	-3.124
<i>BBB+ × Year 2005 (Year 2005)</i>	-0.262	-0.580	-0.496	-0.977	-1.351**	-2.394
<i>BBB+ × Year 2006 (Year 2006)</i>	-2.105***	-2.679	-2.293***	-2.768	-3.289***	-3.469
<i>BBB+ × Year 2007 (Year 2007)</i>	-2.325***	-2.975	-2.526***	-3.090	-3.421***	-3.883
<i>BBB+ × Year 2008 (Year 2008)</i>	-1.202**	-2.295	-0.915	-1.529	-1.807***	-3.164
<i>BBB+ × Year 2009 (Year 2009)</i>	-2.387**	-2.203	-2.671**	-2.375	-3.271***	-2.883
<i>BBB+ × Year 2010 (Year 2010)</i>	-0.724	-1.494	-0.778	-1.472	-1.073**	-2.102
<i>BBB+ × Year 2011 (Year 2011)</i>	-1.133**	-2.091	-1.316**	-2.120	-1.912***	-3.129
<i>BBB+ × Year 2012 (Year 2012)</i>	-1.069**	-2.000	-1.136*	-1.913	-1.450***	-2.669
<i>BBB+ × Year 2014 (Year 2014)</i>	-1.682**	-2.505	-1.559**	-2.195	-2.251***	-3.363
<i>BBB+ × Year 2015 (Year 2015)</i>	-1.987**	-2.513	-1.750**	-2.079	-2.459***	-2.864
<i>BBB+ × Year 2016 (Year 2016)</i>	-1.352**	-2.215	-1.453**	-2.251	-1.778***	-2.820
<i>BBB+</i>	0.564*	1.933	0.669**	1.965		
<i>EDF</i>	0.113	0.617	0.260	0.902	0.274	0.414
<i>DEDF</i>	-0.760***	-4.883	-0.312	-0.629	-1.058	-1.176
<i>Constant</i>	-1.243***	-4.496	-3.568***	-6.015	-4.454***	-2.859
Observations	11,865		4545		1079	
Controls	Yes		Yes		Yes	
Pseudo R ²	0.0797		0.0920		0.227	
Panel B: More Recent Period						
Variables	(1)		(2)		(3)	
	Coef.	z-stat	Coef.	z-stat	Coef.	z-stat
<i>BBB+ × Year 2011 (Year 2011)</i>	-1.433***	-2.805	-1.671***	-2.822	-2.092***	-3.310
<i>BBB+ × Year 2012 (Year 2012)</i>	-0.797*	-1.840	-0.766	-1.532	-1.084**	-2.542
<i>BBB+ × Year 2014 (Year 2014)</i>	-1.551***	-2.864	-1.321**	-2.292	-1.751***	-3.105
<i>BBB+ × Year 2015 (Year 2015)</i>	-0.772*	-1.738	-0.703	-1.398	-1.175***	-2.699
<i>BBB+ × Year 2016 (Year 2016)</i>	-1.211**	-2.395	-1.502***	-2.782	-1.369***	-2.689
<i>Constant</i>	-0.782**	-2.341	-2.715***	-4.040	-1.930	-1.230
Observations	6567		2739		674	
Controls	Yes		Yes		Yes	
Pseudo R ²	0.0637		0.0757		0.163	
Panel C: Downgrades of A- Firms						
Variables	(1)		(2)		(3)	
	Coef.	z-stat	Coef.	z-stat	Coef.	z-stat
<i>A- × Year 2003 (Year 2003)</i>	-0.273	-0.385	-0.468	-0.623	0.275	0.353
<i>A- × Year 2004 (Year 2004)</i>	-0.230	-0.318	0.049	0.064	0.261	0.351
<i>A- × Year 2005 (Year 2005)</i>	0.832	1.326	0.940	1.393	1.680**	2.440
<i>A- × Year 2006 (Year 2006)</i>	-0.309	-0.432	-0.228	-0.301	0.320	0.424
<i>A- × Year 2007 (Year 2007)</i>	0.427	0.626	0.466	0.647	1.000	1.440
<i>A- × Year 2008 (Year 2008)</i>	0.202	0.293	0.447	0.620	0.768	1.095
<i>A- × Year 2009 (Year 2009)</i>	1.039	1.613	1.321*	1.902	1.630**	2.347
<i>A- × Year 2010 (Year 2010)</i>	-0.621	-0.672	-0.739	-0.765	0.040	0.042
<i>A- × Year 2011 (Year 2011)</i>	0.663	0.940	1.156	1.511	0.979	1.348
<i>A- × Year 2012 (Year 2012)</i>	0.835	1.256	1.040	1.423	1.180*	1.740
<i>A- × Year 2014 (Year 2014)</i>	-0.742	-0.820	-0.719	-0.756	-0.777	-0.857
<i>A- × Year 2015 (Year 2015)</i>	0.293	0.447	0.308	0.434	0.689	0.997
<i>A- × Year 2016 (Year 2016)</i>	0.068	0.105	0.446	0.633	0.717	1.083
<i>A-</i>	0.118	0.218	0.032	0.057		
<i>Constant</i>	-3.724***	-12.559	-3.982***	-6.706	-4.268***	-3.522
Observations	15,185		5393		981	
Controls	Yes		Yes		Yes	
Pseudo R ²	0.104		0.108		0.186	

Panel A of this table reports the results of regressions in Table 4 with expected default forecasts controlled for. Panel B reports the results of regression in Table 4 with the sample for 2011–2016. Panel C presents the results of regressions that are the same with those in Table 4 with the dependent variable being the indicator for downgrades and the key independent variable being the interaction of the indicator variable for A- firms and Year 2013. Control variables are the same with those in the main tests. See Appendix C for variable definitions. Standard errors are clustered at the firm level. ***, ** and * denote statistical significance at the 1%, 5% and 10% levels, respectively.

conditional on the differences in ratings between S&P and EJ R in other years, the S&P ratings for its A- firms were higher than EJ R ratings in 2013. In contrast, the coefficient on *A- × Year 2013* is positively significant, and neither of the coefficients on *BBB+ × Year 2013* or *BBB × Year 2013* is statistically significant, which suggests that for firms with ratings adjacent to A-, S&P's ratings were either lower or indifferent

from EJ R ratings in 2013. The results are consistent with that S&P lowers the bar for A- firms in 2013, even compared with EJ R.

7. Robustness tests

We further conduct several robustness tests to consolidate our

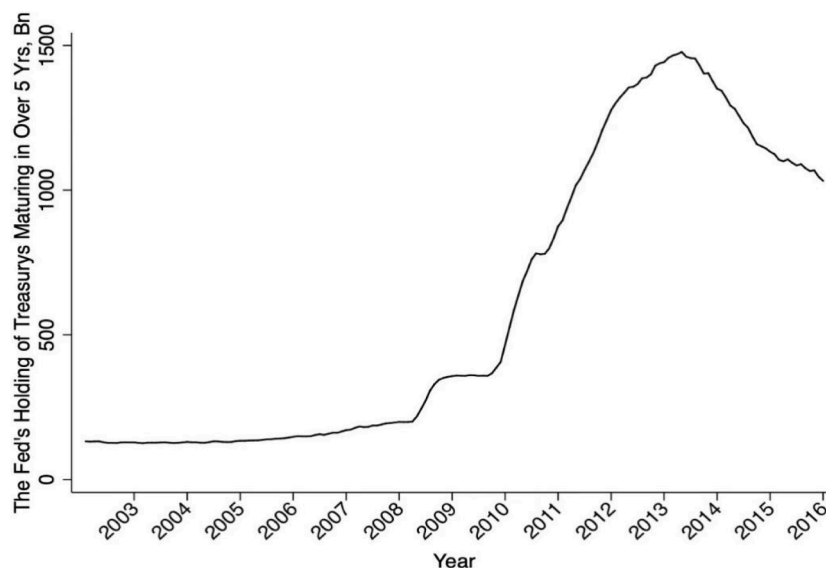


Fig. 2. Long-term treasury holdings by the federal reserve.

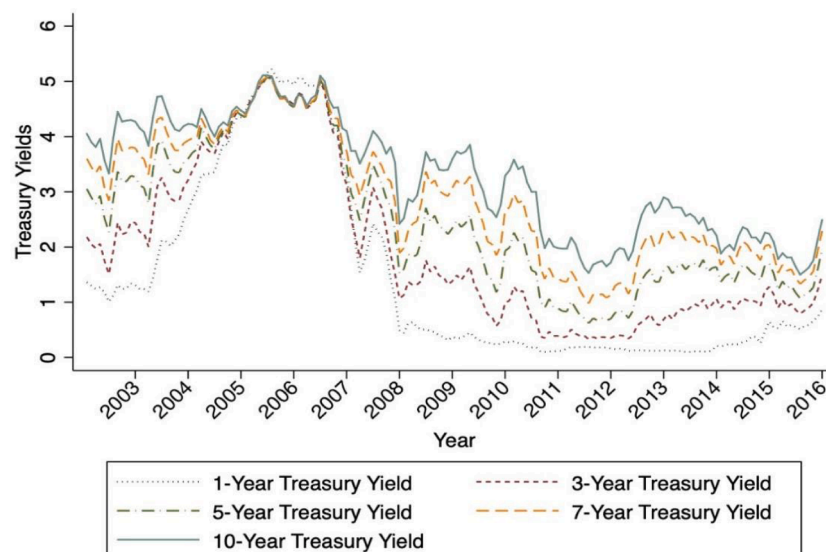


Fig. 3. Treasury yields.

conclusion. First, in the main tests, we include a number of variables to control for firms' financial performance and also perform tests to exclude the possibility that our results are driven by macroeconomic shocks that are common to all firms with certain ratings. But it is still possible that our results are driven by macroeconomic shocks that only affect the default risks of BBB+ firms. To exclude this possibility, we add firms' expected default forecast (EDF) developed by Merton (1974) and its changes to the tests reported in Table 4 to further control for firms' actual credit risks. EDF is calculated based on firms' stock returns and reflects investors' overall evaluation of firms' risks. As presented in Panel A of Table 11, our results do not change after including EDF and its changes.

Moreover, our sample covers a long period (2003–2016), but as Alp (2013) indicates, there could be a structural change in CRAs' credit ratings. To mitigate concerns over the impact of such structural changes on our results, we perform the tests in Table 4 again using a subsample covering a shorter period (2011–2016). As shown in Panel B of Table 11, the results remain similar to those in Table 4.

Additionally, we examine whether our results are driven by S&P

downgrading more A- firms in 2013. Specifically, we perform tests similar to those in Table 4 with the dependent variable replaced with an indicator variable for downgrades. The results in Panel C of Table 11 suggest that there were no abnormally more downgrades for A- firms in 2013.²⁴

8. Conclusion

Prior studies have shown that rating-based regulations often induce rating inflations. Exploiting the surge in demand for A- bonds after the Fed's large-scale asset purchase programs, we contribute to this strand of literature by providing new evidence on how rating-based regulations affect rating outcomes. We show that the S&P, an issuer-paid rating agency, upgraded abnormally more BBB+ firms to the A- rating category

²⁴ In an untabulated test, we also define firms on negative watchlist or with negative outlook as being downgraded, and there were still no abnormal downgrades from A- to BBB+ in 2013.

when the regulatory advantage of A- securities was larger, but the upgraded firms did not subsequently exhibit relative improvement in financial performance. After the upgrades, instead of making more investments, the upgraded BBB+ firms increased their payouts to investors. Furthermore, it was the firms with a higher interest burden (i.e., larger potential economic benefits from upgrades) that were more likely to be upgraded when the cost of debt was relatively low. These firms had also issued more bonds prior to the abnormal upgrades and subsequently issued more bonds. Overall, our results highlight the time-varying nature of rating-based regulations. Our findings also suggest that CRAs still possess significant discretion in assigning ratings after the passage of the Dodd-Frank Act. Finally, this study contributes to the recent literature on rating inflations during the quantitative easing programs (Acharya

et al., 2021; Abidi et al., 2023) and complements the picture portrayed by Foley-Fisher et al. (2016) about the consequences of the Fed's unconventional monetary policies.

CRediT authorship contribution statement

Yao Tang: Writing – review & editing, Writing – original draft, Visualization, Conceptualization. **Xiao Xiao:** Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Software, Resources, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization.

Appendix A

Capital Requirements for Bonds Held by Life Insurance Companies

NAIC Category	Bond Ratings	Capital Charge (before tax)	Capital Charge (after tax)
US Government Bonds		0	0
Risk Category 1	AAA, AA+, AA, AA-, A+, A, A-	0.40%	0.30%
Risk Category 2	BBB+, BBB, BBB-	1.30%	1.00%
Risk Category 3	BB+, BB, BB-	4.60%	3.40%
Risk Category 4	B+, B, B-	10.00%	7.40%
Risk Category 5	CCC+, CCC, CCC-	23.00%	17.00%
Risk Category 6	lower and non-rated	30.00%	19.50%

Source: The American Academy of Actuaries C1 Work Group²⁵ and the Pacific Investment Management Co.²⁶.

Appendix B

Quantitative Easing and Demand for A- Debt Securities

The aftermath of the 2008–2009 global financial crisis forced many countries to adopt aggressive monetary and fiscal easing. In the US, the federal government rolled out a \$783- billion stimulus plan as a fiscal response. On the monetary side, the Fed promptly cut the federal funds rate to zero to support the economy. The dire economic conditions during and after the crisis made it clear that cutting the interest rate to zero was not sufficient to bolster the economy (i.e., a situation often referred to as the zero-lower-bound problem). In this situation, conventional monetary policies which feature interest rates at their center were exhausted. As a solution to the zero-lower-bound problem, the Fed and a few other major central banks employed QE, an unconventional monetary policy. The idea is to inject liquidity in the form of base money into the financial sector.

To be specific, the Fed conducted three rounds of quantitative easing between 2008 and 2014. In the first round of QE, the Fed printed \$1 trillion and \$650 billion to purchase US government bonds between December 2008 and March 2010. In the second round of QE, implemented between November 2010 and June 2011, the Fed targeted long-term interest rates to boost borrowing in the private sector. To achieve the goal, the Fed purchased \$600 billion worth of government bonds with longer maturity.

Between September 2011 and December 2012, the Fed implemented the MEP and purchased \$667 billion long-term Treasury securities. The third round of QE is the most aggressive because the scale of asset purchase was not predetermined. Rather, starting from September 2012, the Fed was committed to purchasing \$40 billion worth of mortgage-backed securities (MBS) per month until it deemed that the labor market had improved significantly in the context of price stability. In December 2012, the pace of asset purchases was increased to \$85 billion per month, which included \$45 billion worth of long-term government bonds and \$40 billion worth of MBS. As economic conditions gradually improved, the third round of QE started to taper off in February 2014 (i.e. the pace of asset purchase started to decrease). Eventually, this round of QE came to a complete stop in October 2014.

As shown in Fig. 2, the Fed's holding of long-term Treasury securities with maturities over five years kept increasing since 2007 and peaked around the end of 2013. The Fed's holding of long-term Treasury securities was rising quickly through 2012 and 2013, registering the highest rates of increase at 70.5% and 50.3% during the expansion of the Fed's balance sheet.

The strong demand for long-term Treasury securities during MEP and QE3 pulled down the yields of Treasury securities and caused investors who

²⁵ See <https://www.actuary.org/sites/default/files/files/imce/Academy%20C1WG%20Documentation%20Corp%20Bond%20Factors%20%20Aug%203%202015%20Final.pdf>

²⁶ See <https://www.pimco.com/en-us/insights/viewpoints/research/proposed-changes-in-risk-based-capital-rules-for-us-life-insurance-investments>

strongly preferred low-risk fixed-income securities to switch to high-quality corporate bonds for higher yields. The demand was especially strong for A- bonds because insurance companies, an important type of large institutional investors, were subject to risk-adjusted capital requirements based on credit ratings, and A- bonds provided the highest yields among corporate bonds that cost the lowest level of capital (Foley-Fisher et al., 2016).²⁷ Insurance companies' holdings of bonds in Category 1 (i.e., bonds with ratings between AAA+ and A-) represented 66.6% of the industry's total bonds as of year-end 2016. Insurance companies hold approximately 38% of all outstanding U.S. corporate bonds (Federal Reserve Financial Accounts 2017) with around 80% of their corporate bond holdings rated between A and Baa (Becker and Ivashina 2015; NAIC Capital Markets Bureau 2023).

Another unconventional monetary policy adopted after the 2008 crisis, forward guidance, also contributed to the strong demand for A- bonds. In the forward guidance in 2013, the Fed essentially assured the general public that the federal funds rate would remain low in the future. Such expectation of a low-interest rate environment undoubtedly pushed up the demand for corporate bonds, especially the demand for A- bonds from insurance companies.

Fig. 3 presents the yields of long-term Treasury securities for the period of 2003–2016. Due to the combined effects of MEP, the third round of QE, and forward guidance, yields of Treasury securities remained stably low between mid-2011 and mid-2013. Such drops in yields of Treasury securities no doubt added to the appeal of A- corporate bonds. Although Treasury yields started to rise in May 2013 due to changes in expectation about the tapering off and exit from QE, it fell in the September of the same year and was still low compared to those in the ex- and post-QE periods.²⁸

Fig. 1 plots the median yield spreads (i.e., premium relative to Treasury securities) of A, A-, and BBB+ bonds for 2011–2015. The yield spreads of these bonds declined in 2013 and the spread between A and A- bonds (i.e., the difference in yield spreads between these bonds) narrowed in the same period, which is consistent with a strong demand for A- bonds. Although the spread between A and A- bonds continued to decrease in 2014, their yield spreads against Treasury securities went up, which suggests the demand for A- bonds weakened. The spread between A- and BBB+ bonds in 2012–2013 were higher than that of the other periods, which is consistent with that the significant rise in regulation-induced demand for A- bonds increased the regulatory advantage of A- bonds relative to BBB+ bonds in this period.

Appendix C

Variable Definitions

<i>AbUP_Client</i>	=	An indicator variable equal to one if a previously identified borrower with debt capacity was abnormally upgraded (by one notch) in 2013, and zero otherwise.
<i>AbUP_Client2</i>	=	An indicator variable equal to one if a previously identified borrower with debt capacity was abnormally upgraded (by two notches) in 2013, and zero otherwise.
<i>Active_Client</i>	=	An indicator variable equal to one if a firm issued bonds in the three years prior to its upgrades but the amounts were less than the median level.
<i>Cash</i>	=	Cash and marketable securities divided by total assets.
<i>CFODebt</i>	=	Cash flow from operations divided by total debt.
<i>Convert</i>	=	Convertible debt divided by total assets.
<i>Debt</i>	=	The sum of long-term and short-term debt divided by EBITDA.
<i>DX</i>	=	The change of control variable X.
<i>EDF</i>	=	Expected default forecast constructed by Merton (1974).
<i>HighDemand</i>	=	An indicator variable equal to one for years between 2012 and 2013, and zero otherwise.
<i>High_Debt</i>	=	An indicator variable equal to one if a firm's <i>Debt</i> for a year is higher than the median level for the year and the same rating category, and zero otherwise.
<i>High_IntCov</i>	=	An indicator variable equal to one if a firm's <i>IntCov</i> for a year is higher than the median level for the year and the same rating category, and zero otherwise.
<i>High_Int/TDebt</i>	=	An indicator variable equal to one if a firm's interest expenses divided by total debt for a year is higher than the median level for the year and the same rating category, and zero otherwise.
<i>IntCov</i>	=	Interest payments divided by EBITDA.
<i>Leverage</i>	=	The sum of long-term and short-term debt divided by total assets.
<i>Post</i>	=	An indicator variable equal to one if for years between 2013 and 2015, and zero otherwise.
<i>ProfitMargin</i>	=	EBITDA divided by sales.
<i>PPE</i>	=	Net property, plant, and equipment divided by total assets.
<i>Predown</i>	=	An indicator variable equal to one if a firm was downgraded between 2007 and 2009, and zero otherwise.
<i>Profit</i>	=	EBITDA divided by sales.
<i>ProVol</i>	=	The standard deviation of Profit over the past five years.
<i>Rent</i>	=	Rental payments divided by total assets.
<i>ROA</i>	=	EBITDA divided by total assets.
<i>Size</i>	=	The logarithm of the book value of assets.
<i>TrendTd</i>	=	The coefficient on time trend from regressing total debt on time for the previous eight quarters ending by the end of the previous year, multiplied by 100.
<i>TrendDlt</i>	=	The coefficient on time trend from regressing total long-term debt on time for the previous eight quarters ending by the end of the previous year, multiplied by 100.
<i>TrendROA</i>	=	The coefficient on time trend from regressing ROA on time for the previous eight quarters ending by the end of the previous year, multiplied by 100.
<i>TrendIntcov</i>	=	The coefficient on time trend from regressing interest coverage on time for the previous eight quarters ending by the end of the previous year.
<i>UP</i>	=	An indicator variable equal to one if a firm is upgraded in the current year, and zero otherwise.
<i>UP2013</i>	=	An indicator variable equal to one if a BBB+ firm was upgraded in 2013, and zero otherwise.
<i>UP_Abnormal</i>	=	An indicator variable equal to one if a firm is upgraded to A- but the estimated rating score is greater than the actual rating score by one notch.
<i>UP_Abnormal2</i>	=	An indicator variable equal to one if a firm is upgraded to A- but the estimated rating score is greater than the actual rating score by two notches.
<i>Year T</i>	=	Indicator variables for year T.

²⁷ Becker and Ivashina (2015) examine the investment behavior of insurance companies, the largest class of institutional investors in bonds. Because regulations require insurance companies to hold more capital when they invest in bonds with lower credit ratings (see Appendix A for more details on the rating-based capital requirements), they find that insurers prefer to hold bonds with higher ratings. Interestingly, Becker and Ivashina (2015) also find that within the same risk category, insurance companies would reach for yield by picking bonds with higher yields and higher ex-post risk, which is referred to as "reaching-for yield". Consistent with the finding in Becker and Ivashina (2015), Foley-Fisher et al. (2016) provide evidence that the yields of A- bonds were significantly lower and insurance companies held a remarkably higher proportion of A- bonds in the period of the MEP.

²⁸ In Appendix D, we provide detailed explanations for the fluctuation in Treasury yields in 2013.

Appendix D

Causes of the Fluctuations in Treasury Yields in 2013

Three events are crucial for understanding the ups and downs of Treasury yields between May 2013 and September 2013. On May 22, 2013, the former Chair of the Fed Ben Bernanke suggested in his testimony at the US Congress that the Fed would slow down the pace of QE “in the next few meetings”, prompting the financial market to revise downward the expectation of continued monetary easing.²⁹ Consequently, the yields of Treasury securities began to rise. On June 19, 2013, the economic projection data released by the Federal Open Market Committee (i.e., the committee responsible for setting monetary policy within the Fed system) indicated an improvement in expected economic performance in 2014, further strengthening the market’s suspicion that the tapering and eventual end of QE would be close.³⁰ The market expected that the tapering of QE could be announced as early as in September 2013, hence the Treasury yields hiked sharply and climbed in the August of 2013.³¹ The FOMC statement on September 18, 2013, however, delivered a major surprise to believers of QE tapering by maintaining the pace of QE.³² Treasury yields dropped and remained relatively stable in the subsequent months.

Data availability

Data will be made available on request.

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²⁹ See <https://www.bloomberg.com/news/articles/2013-05-22/treasuries-extend-gains-as-bernanke-says-stimulus-still-needed>.

³⁰ See <https://www.federalreserve.gov/monetarypolicy/files/FOMC20130619SEPcompilation.pdf>.

³¹ See <https://www.cnbc.com/id/100828661>.

³² See <https://www.nytimes.com/2013/09/13/business/economy/fed-prepares-for-change-in-policy-and-in-policy-makers.html>, and <https://www.usatoday.com/story/money/business/2013/09/18/sept-federal-reserve-meeting/2828859/>.